CN Practical 5: Create a Star Topology Network

Aim: To create a star topology network by connecting computers using wired and wireless media.

Theory:

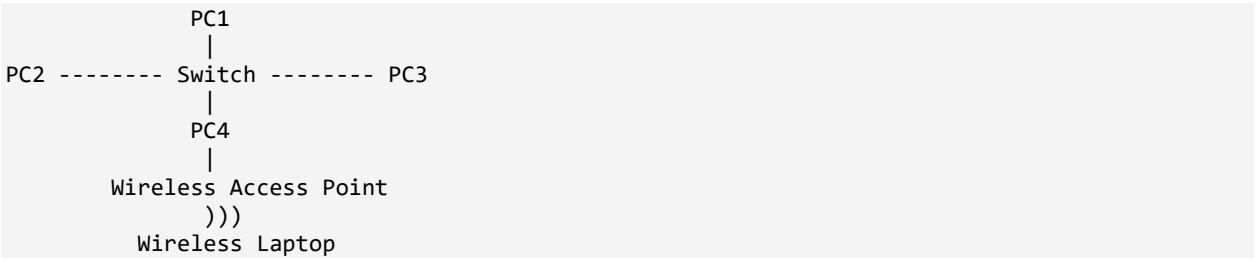
Star topology is a network arrangement in which all computers and devices connect to a central networking device, usually a switch or wireless access point. In a wired star network, each computer has an individual Ethernet cable connected to the switch. Wireless devices may connect through a wireless access point integrated with a router or connected to the same LAN.

Star topology is widely used because it is easy to install, expand, manage and troubleshoot. If the cable connecting one computer fails, the other computers can continue to communicate. However, if the central switch fails, the devices connected through it may lose network connectivity. Star topology is suitable for computer laboratories, offices and educational institutions.

Materials required:

- Computers or laptops
- Ethernet switch
- Ethernet cables
- Wireless router or access point
- Network interface cards

Network diagram:



Procedure:

1. Open the required networking environment or arrange the physical devices.
2. Place the switch at the centre of the network.
3. Connect the wired computers to the switch using Ethernet cables.
4. Connect a wireless access point to the LAN if wireless connectivity is required.
5. Enable Wi-Fi on the laptop and connect it to the access point.
6. Assign unique IP addresses to the computers in the same subnet, or use DHCP.
7. Verify the physical connections and ensure the network interfaces are enabled.
8. Use the ipconfig command to check the IP addresses.
9. Use the ping command to test communication between the computers.
10. Observe how the devices communicate through the central networking equipment.

Observation:

All wired computers were connected to a central switch, while the wireless laptop communicated through the access point. The devices were configured on the same network, and connectivity was tested using the

ping command. The failure of an individual endpoint cable would normally affect only that endpoint, while failure of the central switch could disrupt the wired network.

Conclusion:

A star topology network was created using wired and wireless connections. The practical demonstrated the role of the central switch and wireless access point in network communication. The advantages and limitations of star topology were studied successfully.